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1. Introduction to ALMA

1.1. Science & Capabilities

The Atacama Large Millimeter/submillimeter Array (ALMA) is currently the largest radio interferometer, consisting of 66 antennas with diameters of 12 m and 7 m, located at 5050 m on the Chajnantor Plateau in the Atacama Desert of Chile. With antenna separations (or baselines) of up to about 16 km, ALMA can reach exceptional angular resolutions of a few milliarcseconds ($0''.005$ at 950 GHz). ALMA operates across ten high-frequency receiver bands, spanning 35 to 950 GHz and covering the mm and sub-mm regimes (see Fig. 1).

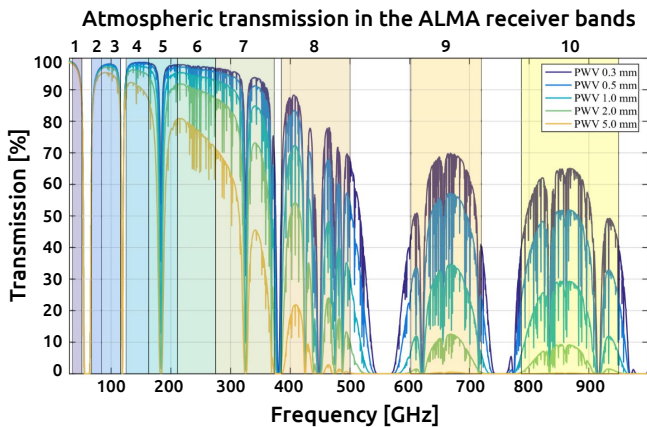


Figure 1: Atmospheric transmission across the ten ALMA receiver bands (from [Carpenter et al. 2020](#)). The curves correspond to different amounts of precipitable water vapor (PWV), which is the total amount of water vapor integrated along the atmospheric column. ALMA bands are placed in the atmospheric windows where transmission is highest. At higher frequencies, especially Bands 8 to 10, very dry conditions become essential for the observations.

ALMA is an extremely versatile facility, providing high-fidelity imaging, sensitive spectroscopy to study molecular gas and chemistry, and continuum observations to trace dust and star formation. It also offers polarization capabilities to probe magnetic fields, and can be used in VLBI mode to reach event-horizon-scale resolution, as demonstrated by the Event Horizon Telescope (e.g., [EHT Collaboration et al. 2019](#)).

With the upcoming Wideband Sensitivity Upgrade (WSU, [Carpenter et al. 2022](#)), ALMA will enter a new phase of enhanced sensitivity, increased observing speed, and greater

calibration flexibility, making this an ideal time to learn how to design observing programs and process ALMA data.

1.2. ALMA Observations & Data

Because sky frequencies are too high to be processed directly by the receiver electronics, the sky signal is first converted to a lower frequency using a mixer and a very stable reference frequency called local oscillator (LO). As a result, most of the ALMA bands are heterodyne receivers with two sidebands (see Fig. 2). Band 1 is the only single-sideband receiver.

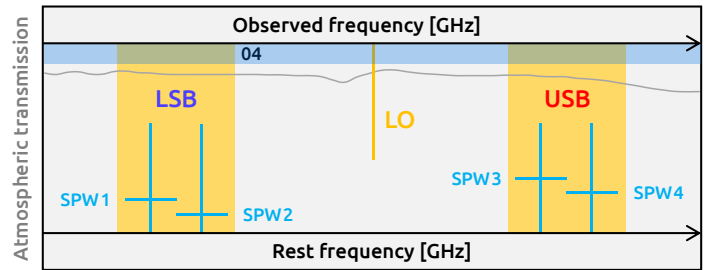


Figure 2: Illustration of a spectral setup using the ALMA Band 4 receiver in the Observing Tool (OT). The observed frequency axis is split into a lower sideband (LSB) and an upper sideband (USB) around the local oscillator (LO), a stable reference frequency. Four spectral windows (SPWs) are distributed across the two sidebands. The solid gray curve represents the atmospheric transmission.

ALMA observations can use up to four spectral windows (SPWs), distributed between the lower sideband (LSB) and upper sideband (USB) around the LO frequency (see Figs. 2 and 3), enabling simultaneous continuum and line imaging. Each SPW is processed by an ALMA antenna correlator and can be configured in Time Division Mode or Frequency Division Mode, the latter providing high spectral resolution.

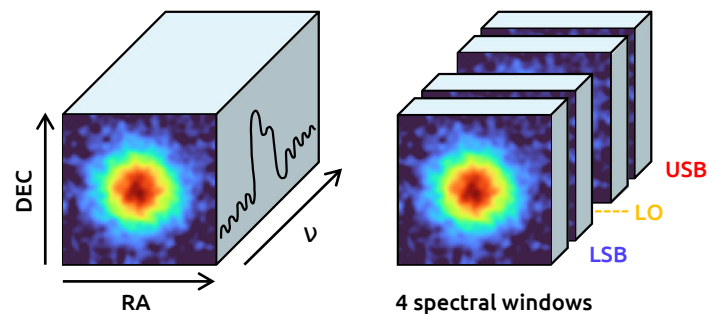


Figure 3: *Left:* Representation of a mock ALMA datacube with two spatial angular dimensions (RA, Dec) and one spectral dimension (frequency axis ν), with the emission spectrum illustrated. *Right:* Four datacubes corresponding to four spectral windows, distributed between the lower sideband (LSB) and upper sideband (USB) around the local oscillator (LO) frequency.

⇒ Further details on ALMA capabilities can be found [here](#).
 ⇒ The following sections present concise notes introducing the main steps of ALMA data processing using [CASA](#).

2. Getting Started with CASA

During an observing cycle, CASA and the ALMA Pipeline are regularly updated. To reduce a specific dataset, one should use the [CASA version](#) (.tar.xz or .dmg file) associated with it, which can be found in the Quality Assurance 2 (QA2) report in [SnooPI](#) or in the [ALMA Archive](#). Once extracted, you can create an alias to launch CASA from any directory:

```
>>> nano ~/.bashrc
Add a new alias, e.g., casa661="/path/to/your/casa"
# casa661 ≡ alias for launching CASA 6.6.1
>>> source ~/.bashrc
>>> casa661
```

Note: The CASA version associated with the observing cycle is needed to recover the calibrated uv products (Sect. 4) or for self-calibration. For imaging (Sects. 5-12), it is recommended to use the latest CASA version, which is often the most stable.

3. ALMA Data Retrieval

The [ALMA Science Archive](#) (see Fig. 4) contains all data obtained with ALMA for science purposes and is accessible to all users after a 1-year proprietary period. Scheduled projects not yet observed can be found on the [Priority Projects](#) list.

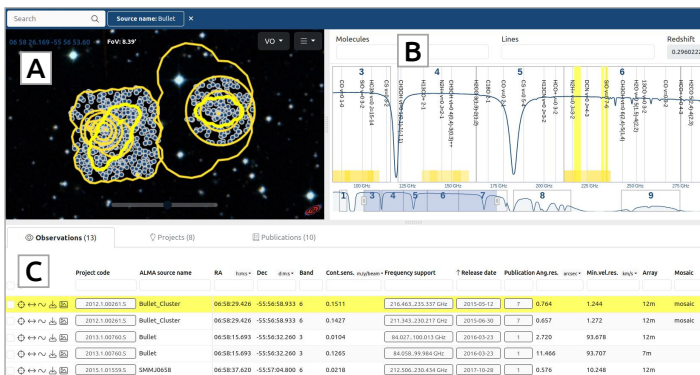


Figure 4: ALMA Science Archive interface. **A:** Sky view of the ALMA programs for a query for the Bullet Cluster. **B:** Spectral line coverage across the receiver bands. One can search for any spectral transition at the source redshift. **C:** List of available observations and their metadata (project ID, target, array, band, resolution, etc.).

After launching a query on the archive, select a dataset, click *Download*, then search for the raw (.asdm.sdm.tar) and auxiliary (.auxiliary.tar) products. Alternatively, click *Open legacy Request Handler* (where all scheduling blocks are mixed), tick the *raw* and *auxiliary* boxes, deselect any irrelevant data, then click *Download Selected* and *Download Script*. Run the script in the desired directory using:

```
>>> chmod u+x download*.sh
>>> ./download*.sh
```

4. Restoring Calibrated uv -Products

After downloading the data from the [ALMA Archive](#) and the correct [CASA](#) version for running the cycle-related pipeline, go to the directory where the script for the PI is stored, which is needed to restore the calibrated uv products:

```
>>> cd 2026.1.00471.S/science_goal[.]/group[.]/
      member[.]/script
# [.] ≡ uid code
```

Then launch CASA specifying the pipeline is run:

```
>>> casa676 --pipeline
# casa676 ≡ alias for launching CASA 6.7.6
>>> SPACESAVING=2
# Parameter that controls the use of disk space
# during data processing.
>>> execfile("ScriptForPI.py")
# Script to generate calibrated uv products.
```

Once the process is complete, the calibrated uv products are stored in the [calibrated](#) directory (.ms.split.cal files).

5. Image Properties

The [LISTOBS](#) task provides all the essential data information for imaging, which is stored in the measurement sets (.ms).

```
>>> listobs(vis=[filename].ms.split.cal)
# vis ≡ visibilities are uv-plane measurements
```

To begin, we will determine the imaging parameters [imsize](#) and [cell](#), which are specific to each ALMA observation.

The primary beam (FOV), θ_{PB} , can be computed as

$$\theta_{PB} ['] = \theta_{ref} \times \frac{\nu_{ref}}{\nu_{obs}}, \quad (1)$$

where $\theta_{ref} = 19''$ ($33''$) at $\nu_{ref} = 300$ GHz (or $\lambda_{ref} = 1.0$ mm) are the reference values for the 12-m (7-m) array¹.

The synthesized beam (PSF), θ_{res} , can be computed as

$$\theta_{res} ['] = \frac{76}{B_{max} [km] \times \nu_{obs} [GHz]}, \quad (2)$$

where B_{max} is the maximum separation between two antennas in the array configuration (i.e., the maximum baseline), and ν_{obs} is the observing frequency of the representative spectral window of the observations. B_{max} and ν_{obs} can be found in the QA2 report (in [SnooPI](#), or in the downloaded data in [auxiliary > qa](#)). A common practice is to sample the spatial scale with at least five pixels by setting a cell size of $\theta_{res}/5$. The image size in pixels is about $2 \times \theta_{PB} / \text{cell size}$. For large θ_{PB} values, one can use a cropped field to speed up imaging.

¹See [ALMA Basics](#).

6. Multi-Frequency Synthesis (M.F.S.) Imaging

The CASA task for imaging is `TCLEAN`. The default M.F.S. imaging mode produces a 2D map [Jy beam^{-1}] that contains both continuum and line emission, if any. It takes the visibilities from all frequency channels and maps them onto a single spatial-frequency grid. A single or multiple measurement sets (or visibility files) can be combined, e.g., for observations taken at different times and/or using different arrays.

In the following example, there are two measurement sets:

```
>>> tclean(vis=["[...]ms.split.cal",
"[...]ms.split.cal"], field=["2","2"],
imagename="MFS_test", imsize=[300],
cell=["0.1arcsec"], phasecenter="J2000
06h58m37.559 -55d57m05.075", specmode="mfs")
```

Then, one can set `specmode="cube"` to generate a 3D cube [$\text{Jy km s}^{-1} \text{beam}^{-1}$] sampling the four SPWs. Blank channels (or slices) in the cube correspond to frequency gaps between the SPWs and/or the two sidebands (see Fig. 3).

Tip: To restore the latest parameters of any task, one can use `tget [task]`. To save them, rename the `[task].last` file.

7. Data Weighting Schemes

The better the uv coverage of ALMA observations, the better the image reconstruction. During imaging, the weighting parameter controls the trade-off between sensitivity and angular resolution by assigning different weights to the measured visibilities according to their position in the uv plane (see Fig. 5).

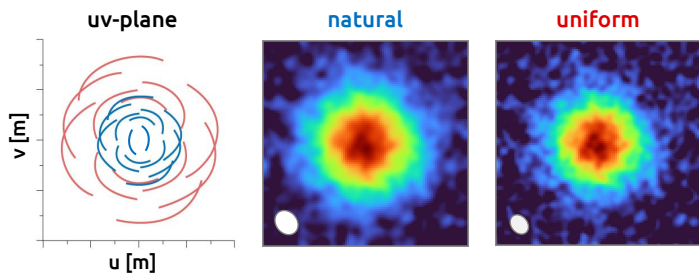


Figure 5: Illustration of uv -plane sampling and two data weighting schemes: the *natural* weighting emphasizes dense uv -plane regions (short baselines), while *uniform* weighting emphasizes extended baselines (outer uv -plane). Ellipses are the corresponding synthesized beams (θ_{res}). The *natural* weighting enhances the S/N at the expense of angular resolution, with the opposite effect for *uniform*. In practice, we often use *Briggs* weighting, which uses a robustness parameter (R) to provide a compromise between the two.

For extended sources, it is recommended to first run `TCLEAN` in interactive mode using `weighting="natural"`. Then, re-run it in non-interactively using the output `.mask` image, and increase the angular resolution by applying Briggs weighting with a robustness parameter, e.g., $R = 0.5$ (common value).

8. Continuum Imaging

We select the spectral channels with no line emission from the 3D cube to produce an (M.F.S.) 2D map [Jy beam^{-1}] that contains only continuum emission. For example:

```
∀ SPW ID ∈ {0,1,2,3}: spw=["0:0~49;101~239,1,2,3"]
```

This means that channels 50 to 100 are excluded from SPW ID 0, which contains a total of 240 spectral channels (IDs 0 to 239). For the other SPWs, all channels are included.

To launch the cleaning process, one should set `niter=10000` (or any sufficiently high number) to allow the `TCLEAN` algorithm to perform enough iterations, and define a threshold at which to stop cleaning. A commonly used threshold is three times the root-mean-square (RMS) noise measured within an emission-free region (CASA Viewer > Region > Statistics). This parameter is set, e.g., as `threshold="3e-5Jy"`.

9. Continuum Subtraction

The best-fit model of continuum emission is subtracted from the uv -plane using the `UVCONTSUB` task. The `spw` parameter specifies the SPWs with line emission, while the `fitspec` parameter tells CASA where to fit the spectrum by specifying the channel ranges where only continuum emission is present. The continuum-subtracted data are generated in the `dir_cal` directory, having the extension `.uvcontsub`. The task must be performed separately for each measurement set.

```
>>> uvcontsub(vis=["[...]ms.split.cal",
field="2", spw="0", datacolumn="data",
fitspec="0:0~49;101~239,1,2,3")
```

10. Spectral Line Imaging

This time, we select spectral channels encompassing the line emission to create a 3D line cube [$\text{Jy km s}^{-1} \text{beam}^{-1}$]. It is recommended to generate a cube with a total velocity width of 2000 km s^{-1} that is centered on the spectral line. One can set `restoringbeam="common"` to ensure that all channels are convolved to the same synthesized beam for consistent spatial resolution across the line cube. As for continuum imaging, we suggest first running `TCLEAN` in interactive mode applying natural weighting, and then in non-interactive mode using the output `.mask` image and Briggs weighting with $R = 0.5$.

```
>>> tclean(vis=["[...]ms.split.cal.uvcontsub",
field="2", spw="0", imagename="CO_CUBE_3xRMS",
specmode="cube", nchan=80, start="-1000km/s",
width="25km/s", restfreq=["91.5GHz"],
deconvolver="hogbom", restoringbeam="common",
weighting="natural", niter=10000,
threshold="3e-5Jy", interactive=True, mask="")
```


11. Line Moment Maps

Moments are 2D maps derived from a 3D line cube by taking statistical moments along the frequency (velocity) axis. The task used to generate moment maps is [IMMOMENTS](#).

A statistical moment of the line emission is computed as

$$M_n(x, y) = \int v^n S(x, y, v) dv, \quad (3)$$

where $S(x, y, v)$ is the flux density at position (x, y) and line-of-sight velocity v , and n is the moment order.

Moment 0 is the velocity-integrated flux [$\text{Jy beam}^{-1} \text{ km s}^{-1}$], providing the spatial distribution of line emission within the source. We select only the channels containing line emission using `chans`, and include all the pixels with `includepix=-1`. For example:

```
>>> immoments(imagename="[...].image",
moments=[0], axis="spectral", chans="35~55",
stokes="I", includepix=-1, excludepix=-1,
outfile="[...].image.mom0")
```

Moment 1 is the rotation velocity [km s^{-1}] of the spectral line emission, while moment 2 is its velocity dispersion [km s^{-1}]. The RMS noise per channel must be taken into account to reject noisy data by defining a suitable range with `includepix`. For example:

```
>>> immoments(imagename="[...].image",
moments=[1], axis="spectral", chans="35~55",
stokes="I", includepix=[0.0005, 1e10],
excludepix=-1, outfile="[...].image.mom1")
```

Figure 6 shows simulations of the first three moment maps of the mock ALMA line cube shown in Figure 3. The moment maps share the same averaged synthesized beam as the parent line cube from which they are extracted. The blank regions in moments 1 and 2 correspond to the masked (noisy) data.

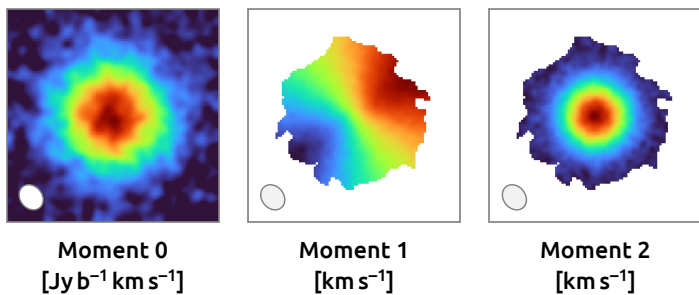


Figure 6: Illustration of the first moment maps of a mock spectral line cube. Moment 0 is the velocity-integrated flux, moment 1 the rotation velocity field, and moment 2 the velocity dispersion.

12. Science-Ready Data Products

12.1. Primary Beam Correction

The primary beam (`.pb` image; Eq. 1) describes the direction-dependent sensitivity of each antenna, defining the effective FOV of an interferometric observation, within which the sensitivity decreases with distance from the pointing center. The observed flux density thus needs to be corrected to recover the intrinsic source emission, especially far from the pointing center. In CASA, this can be done by setting `pbcor=True` when running `TCLEAN`, or using the `IMPBCOR` task.

```
>>> impbcor(imagename="[...].image",
pbimage="[...].pb", outfile="[...].image.pbcor")
```

12.2. Converting to FITS

`EXPORTFITS` converts the CASA products into FITS files.

```
>>> exportfits(imagename="[...].image.pbcor",
fitsimage="[...].fits", dropdeg=False)
# dropdeg=True drops all degenerate axes, i.e.,
the 3rd axis (frequency) and 4th axis (stokes).
```

12.3. Converting to Flux Density per Pixel

One can convert the units of a 2D map of continuum emission or velocity-integrated line emission, $M(x, y)$, from flux density per beam to flux density per pixel as

$$\frac{\Sigma(x, y)}{[\text{Jy pix}^{-1}]} = \frac{M(x, y)}{[\text{Jy beam}^{-1}]} \left(\frac{\Omega_{\text{beam}}}{\Omega_{\text{pix}}} \right)^{-1}, \quad (4)$$

where $\Omega_{\text{beam}}/\Omega_{\text{pix}}$ is the number of pixels per beam.

To account for a source inclination of i , one can multiply the $\Sigma(x, y)$ map by $\cos i$. For a high- z source, the physical scale can be recovered by converting the pixel size from arcsec to physical units at the source redshift (e.g., using `arc2parc`).

A FEW USEFUL LINKS

- [GitHub repository](#) with guidelines on ALMA proposal preparation and data reduction.
- Additional [ALMA tutorials](#) from NRAO.
- Full list of [CASA tasks](#) for data processing.
- More ALMA-related [schematic figures](#).

*Please feel free to reach out if you spot any inaccuracies.
Feedback is very welcome.*

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